# Holographic Error Correction

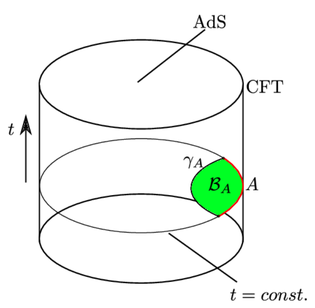
Keshav B. Deoskar [1] Jon D. Goker [1] Mathew J. Thomas [1]

[1]Department of Physics, UC Berkeley

## AdS/CFT Correspondance

Any CFT in $d-$dimensional Spacetime is equivalent to a quantum theory of gravity in a family of spacetimes which are Asymptotically $AdS_d \times M$ for some compact manifold $M$.

## What are "AdS" and "CFT"?

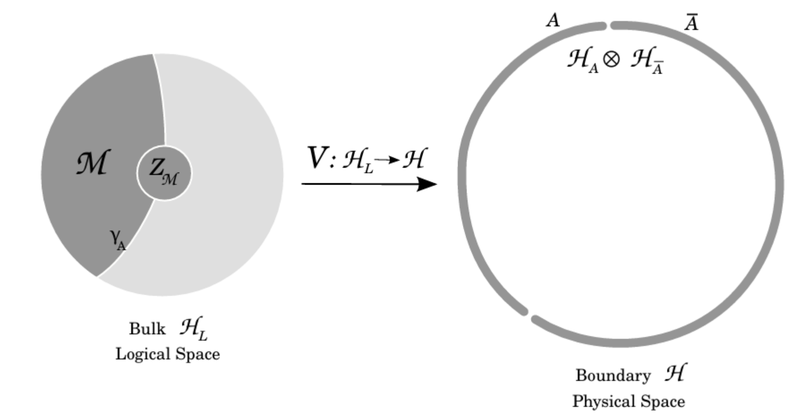



- AdS = Anti-de Sitter Space. Solution to Einstein's Field equations corresponding to spacetime with "negative curvature"
- CFT = Conformal Field Theory. Relativistic QFT which, in addition to Poincaré symmetry is also *scale* invariant $x'^{\mu} \to \lambda x'$

- The AdS/CFT Correspondance gives a dictionary translating between the **bulk gravitational** and **boundary conformal** theories.
- Here, $A$ is a **subregion** of the boundary and $\gamma_A$ is the corresponding **minimal geodesic**.

## What are Holographic Codes? Why are they interesting?

- **Quantum Erasure Correction Codes** that display bulk-boundary correspondance similar to AdS/CFT + properties like Subregion Duality, RT-formula, etc.




[4]

- Provide exactly solvable toy models of AdS/CFT, possibly new perspectives on Topological Order

## Ryu-Takayanagi (RT) Formula

- Generalization of the Bekenstein-Hawking equation

$$S_{BH} = \frac{\text{Area}(\Sigma)}{4G_N}$$

RT formulae relate Entanglement (Quantum Mechanical) to Geodesic Area (Gravitational).

- Original RT Formula,

$$S(\tilde{\rho}_A) = \text{Tr}\left(\tilde{\rho}\mathcal{L}_A\right), \ \mathcal{L}_A = \frac{\text{Area}(\gamma_A)}{4G^{d+2}}$$

- The main result of [2] is:

### Theorem 1.1: Complementary Recovery [2]

For finite-dimensional hilbert space $\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_{\bar{A}}$, code subspace $\mathcal{H}_{code} \subseteq \mathcal{H}$ and Von-Neumann Algebra $M$ acting on $\mathcal{H}_{code}$, the following are equivalent:

- $\exists$ operator $\mathcal{L}_A \in Z_M \equiv M \cap M'$ s.t. $\forall \tilde{\rho}$ on $\mathcal{H}_{code}$ we have

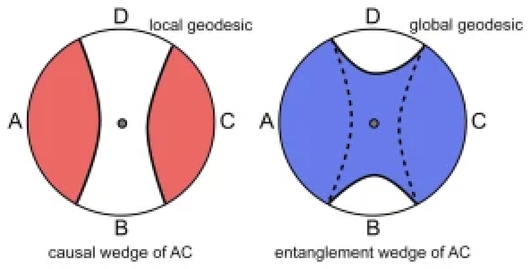
$$S(\tilde{\rho}_A) = \text{Tr}\left(\tilde{\rho}\mathcal{L}_A\right) + S(\tilde{\rho}, M)$$
$$S(\tilde{\rho}_{\bar{A}}) = \text{Tr}\left(\tilde{\rho}\mathcal{L}_{\bar{A}}\right) + S(\tilde{\rho}, M')$$

- For any operators $\widetilde{O} \in M$, $\widetilde{O}' \in M'$ there exist operators $O_A, O_{\bar{A}}$ on $\mathcal{H}_A, \mathcal{H}_{\bar{A}}$ respectively such that for any $|\widetilde{\psi}\rangle \in \mathcal{H}_{code}$ we have

$$O_A|\widetilde{\psi}\rangle = \widetilde{O}|\widetilde{\psi}\rangle, \ O_A^{\dagger}|\widetilde{\psi}\rangle = \widetilde{O}^{\dagger}|\widetilde{\psi}\rangle$$
$$O_{\bar{A}}|\widetilde{\psi}\rangle = \widetilde{O}|\widetilde{\psi}\rangle, \ O_{\bar{A}}^{\dagger}|\widetilde{\psi}\rangle = \widetilde{O}^{\dagger}|\widetilde{\psi}\rangle$$

- So, RT Formula Exists $\iff$ We can reconstruct bulk-operators using $A, \bar{A}$
- [2] shows that every Quantum Erasure Correction Code has an RT-like formula, but the area-operator $\mathcal{L}_A$ is trivial unless we have **Complementary Recovery** and ask for recovery of only a **subalgebra** $M$ of $\mathcal{H}_{code}$.
- So, roughly, a "Good" Holographic Code has non-trivial area operator $\mathcal{L}_A$. More precisely, the condition is that $M$ is **not a factor algebra**.

## Causal and Entanglement Wedges in AdS/CFT



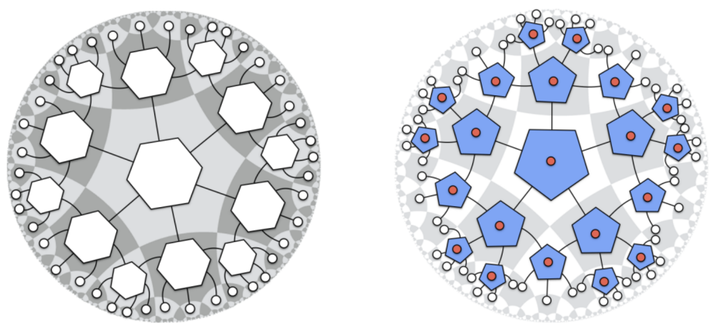

Source: [1]

Associated with a boundary subregion $A$ are:

- **Causal Wedge:** Set of bulk points that can both receive and set signals from/to $A$, denoted $\mathcal{C}[A]$
- **Entanglement Wedge:** Set of bulk points bounded by $A$ and $\gamma_A$ denoted $\mathcal{E}[A]$.
- $\mathcal{C}[A] \subseteq \mathcal{E}[A]$ with $\mathcal{E}[A]$ usually bigger.
- **AdS-Rindler Reconstruction:** procedure for reconstructing bulk-operators using boundary operators.

## Holographic Codes from Tensors: HaPPY Codes

- **Holographic Code:** A tensor network covering a geometric manifold with boundaries that gives rise to an isometric map from uncontracted bulk legs to uncontracted boundary legs. Introduced in [3].
- Uncontracted legs in the bulk / on the boundary, respectively, form logical / physical degrees of freedom.
- There exist other Tensor Network models which display more aspects of AdS/CFT, eg. [5], but these are more complicated.

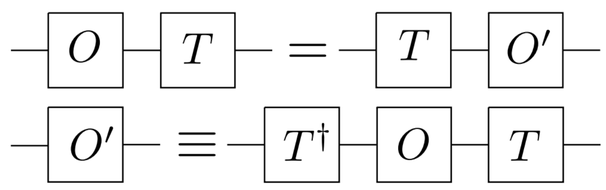


(a) Holographic hexagon state (b) Holographic pentagon code

Two examples of Holographic States and codes respectively

## Perfect Tensors

- A $2n$-index tensor $T_{a_1 \cdots a_{2n}}$ is a **Perfect Tensor** if for any bipartition $A$ of its indices, $T$ is an isometric tensor from $A$ to $A^c$.
- Isometric Tensors have **Operator Pushing** property.




Operator Pushing through Isometric Tensor as Outlined in [3]

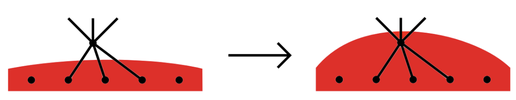
- The simplest examples of a perfect tensor and as such holographic code is the five qubit code, or alternatively 3 qutrit code

## Concrete Example: Application of RT to Tensor Networks

- For instance, for **Pentagon Code** we place 6-index tensors (one left uncontracted in the bulk). The **rate of this code** is shown to be

$$\frac{N_{bulk}}{N_{boundary}} \to \frac{1}{\sqrt{5}} \approx 0.447$$

- For a boundary subregion $A$, we can construct the minimal geodesic $\gamma_A$ with a **greedy algorithm**.



One iteration of the greedy geodesic algorithm

- **Theorem**: *If A is a (not necessarily connected) boundary region and $A^c$ is its complement, then the entropy of A satisfies*

$$S_A \geq \left|\gamma_A^* \cap \gamma_{A^c}^*\right|$$

where $\gamma_A^*$ and $\gamma_{A^c}^*$ refer to the surfaces generated by the greedy geodesic algorithm.

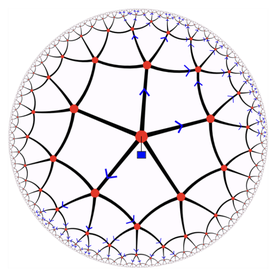
- With these tools in hand, we can properly introduce notions of causal and entanglement wedges in the language of Holographic Codes

## Causal and Entanglement Wedges for HaPPY Codes

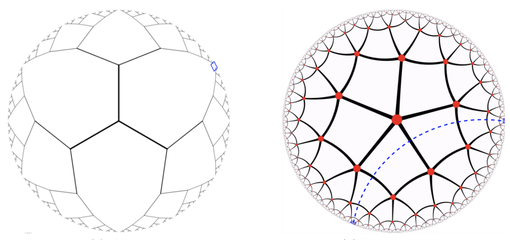
- Define the **"discretized" Causal Wedge** $\mathcal{C}[A]$ for boundary subregion $A$ as set of points reached via the Greedy algorithm.
- Theorem 5 of [3] shows that any bulk operator in $\mathcal{C}[A]$ can be reconstructed as a boundary operator supported on $A$.
- **But we can go further!** Operators *beyond* the causal wedge can be reconstructed. This hints to operators in a larger region, the **(Greedy) Entangelement Wedge** $\mathcal{E}[A]$ being reconstructible. Motivates Entanglement Wedge Reconstruction in full AdS/CFT.

## Error Thresholds

- AdS/CFT has a similar redundancy in its relationship betweem bulk local fields and nonlocal boundary operators, which we can see manifest through tensor pushing



Disconnected reconstruction of bulk operator, where only using the union of the causal wedge of each boundary region can we reconstruct the bulk operator



(a) Triangle code (b) Pentagon code

In the above codes we see a logical error in the flips of 2 and 4 boundary operators respectively

- Error Threshold $p_c$: If probability of erasure $p < p_c$m then $\mathcal{E}[A]$ contains central bulk spin with success probability $1 - \epsilon$ for $\epsilon$ becoming doubly exponentially small as radius of bulk increases.
- **Not all codes have an erasure threshold!** Eg. Pentagon Code. In order for $p_c$ to be defined, code distance must increase with system size.
- For single ubit hexagon code we can obtain a lower bound erasure threshold of

$$p_c \geq \frac{1}{12}$$

, but numerically it can be shown to be quite close to $p_c = 1/2$.

## References

[1] Quantum gravity from quantum error correcting codes? Source: `https://iqim.caltech.edu/2015/03/27/quantum-gravity-from-quantum-error-correcting-codes/`, 2015.

[2] Daniel Harlow. The ryu–takayanagi formula from quantum error correction. *Communications in Mathematical Physics*, 354(3):865–912, May 2017.

[3] Fernando Pastawski, Beni Yoshida, Daniel Harlow, and John Preskill. Holographic quantum error-correcting codes: toy models for the bulk/boundary correspondence. *Journal of High Energy Physics*, 2015(6), June 2015.

[4] Jason Pollack, Patrick Rall, and Andrea Rocchetto. Understanding holographic error correction via unique algebras and atomic examples. *Journal of High Energy Physics*, 2022(6), June 2022.

[5] Matthew Steinberg, Sebastian Feld, and Alexander Jahn. Holographic codes from hyperinvariant tensor networks. *Nature Communications*, 14(1), November 2023.